

PAPER_820: 3D Neutrino-Cooled Accretion Disk Dynamo Cycle UQFF

Unified Quantum Field Framework — Whitepaper 820

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Abstract

This paper integrates the 3D neutrino-cooled accretion disk magnetic dynamo cycle into the Quadriadic UQFF. The 380 ms simulation of a $0.03 M_\odot$ disk around a $3 M_\odot$ BH (spin $\chi = 0.8$) reveals self-sustaining dynamo cycles with $\tau_{dynamo} \approx 20$ ms periodicity driven by the MRI (Magneto-Rotational Instability). The electron fraction self-regulates from midplane $Y_e \approx 0.1$ to outflow $Y_e \approx 0.2$ via neutrino absorption, governing the lanthanide content of early kilonova ejecta. Mass ejection $M_{ej} \approx 0.01 M_\odot$ at $v_\infty \approx 0.1c$ aligns with GW170817 red kilonova constraints.

1. Introduction

The MRI instability drives turbulent angular momentum transport in the accretion disk, with the resulting turbulence generating a self-sustaining dynamo. In the neutrino-cooled regime (relevant for post-merger BH + disk systems at $t < 1$ s), the dynamo cycle operates on timescales comparable to the orbital period, generating cyclically amplifying magnetic fields and periodic mass eruptions.

2. System Parameters

- **Disk mass:** $M_{disk} = 0.03 M_\odot$
 - **Central BH:** $M_{BH} = 3 M_\odot$, spin $\chi = 0.8$
 - **Simulation duration:** 380 ms
 - **Neutrino treatment:** full spectral transport (12 energy bins)
 - **Peak neutrino luminosity:** $L_\nu \approx 10^{52}$ erg/s (initial)
 - **Average:** $L_\nu \approx 10^{51}$ erg/s (integrated over 380 ms)
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3. MRI Timescale

The Magneto-Rotational Instability (MRI) growth rate:

$$\tau_{MRI} = \frac{1}{\Omega}$$

where $\Omega = \left(\frac{GM_{BH}}{r^3} \right)^{1/2}$ is the local orbital frequency. At $r = 10GM/c^2$ for $M_{BH} = 3M_\odot$:

$$\tau_{MRI} \approx 1 \times 10^{-3} \text{ s} = 1 \text{ ms}$$

The dynamo cycle is approximately 20 MRI timescales:

$$\tau_{dynamo} \approx 20 \cdot \tau_{MRI} \approx 20 \text{ ms}$$

UQFF Layer 2:

$$g_{L2,MRI} = \frac{1}{\tau_{MRI}} \approx 10^3 \text{ m/s}^2$$

4. Dynamo Magnetic Field Cycle

The dynamo cycle proceeds as: 1. **Amplification**: MRI amplifies toroidal B_ϕ from seed poloidal B_r 2. **Buoyancy**: Magnetic pressure gradient causes B_ϕ to rise to magnetic corona 3. **Reconnection**: Reconnection at $z \sim H_{disk}$ releases energy, seeds new poloidal B_r 4. **Reset**: Cycle repeats with $\tau_{dynamo} \approx 20 \text{ ms}$

The cycle amplitude:

$$B_{max} \approx \sqrt{4\pi P_{gas}} \approx 10^{15} \text{ G at ISCO}$$

5. Electron Fraction Self-Regulation

The electron fraction (neutron content) evolves under neutrino absorption:

$$\frac{dY_e}{dt} = \dot{Y}_{e,abs} + \dot{Y}_{e,emit}$$

Midplane equilibrium: $Y_e^{eq} \approx 0.1$ (strongly neutron-rich, r -process capable)

Outflow Y_e : rises to ~ 0.2 as material is heated by neutrino absorption during ejection

$$g_{L3,Y_e} = \frac{M_{ej} \cdot Y_e}{r} \approx 2 \times 10^{-7} \text{ m/s}^2$$

6. Dynamo Correction to UQFF

The dynamo cycle enters Layer 4:

$$g_{L4,dynamo} = \frac{\tau_{dynamo}}{r} \approx 2 \times 10^{-6} \text{ m/s}^2$$

This represents the periodic magnetic pressure wave launched at each dynamo reset, propagating through the disk corona at Alfvén speed.

7. Mass Ejection Budget

Total ejecta from 380 ms simulation: - $M_{ej} \approx 0.01M_{\odot}$ ($\approx 40\%$ of initial M_{disk}) - $v_{\infty} \approx 0.1c$ (thermal wind component) - $v_{fast} \approx 0.3c$ (MRI-driven MHD winds) - Average $Y_e \approx 0.2$, containing both lanthanide-rich and lanthanide-poor ejecta

8. GW170817 Red Kilonova Connection

The neutrino-cooled disk wind, with $Y_e \approx 0.2$ and $M_{ej} \approx 0.01M_{\odot}$, matches the GW170817 red kilonova: - Kilonova peak time: $\sim 3-5$ days at visual/NIR wavelengths - Opacity $\kappa \approx 5-10 \text{ cm}^2/\text{g}$ (partial lanthanide enrichment) - Third-peak r -process elements (Xe, Ba, Nd) require $Y_e \leq 0.25$

The dynamo cycle creates periodic neutrino bursts that additionally heat the ejecta, producing optical pulsations at $\tau_{dynamo} \approx 20 \text{ ms}$ (not yet observed but predicted).

9. Summary

The 3D neutrino-cooled disk simulates a 20 ms magnetic dynamo cycle driven by MRI at $\Omega^{-1} \approx \tau_{MRI}$. Electron fraction self-regulates from $Y_e \approx 0.1$ midplane to $Y_e \approx 0.2$ outflow. Mass ejection $M_{ej} \approx 0.01M_{\odot}$ at $v_{\infty} \approx 0.1c$ matches GW170817 red kilonova. These parameters are now registered in the Quadriadic UQFF Layers 2-4.

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